

Supplementary data:

Materials and Methods:

Intracellular Cytokine Staining:

Patient PBMCs from different time-points were thawed, counted, stained with proliferation dye (1 μ l of APC proliferation dye in 1 ml of PBS per 10M) and resuspended at desired concentration. Stimulator cells (either donor or third party splenocytes) were thawed and irradiated at 40 Gy in a facility and stained with proliferation dye (1 μ l of V450 in 1 ml of PBS per 10M). 100,000 responder cells were added to a 96 U bottom-well plate and media, together with 100,000 stimulating cells, media, or CD3/CD28 T cell activator beads (1:1). Plates were incubated at 37°C for 5 days. Following 5 days, cells were harvested into FACS tubes and centrifuged at 400xg for 5 minutes. Cells were then resuspended with 1xBefeldin A in 1 ml of Complete Media (1:1000 dilution) and incubated at 37°C for 4h. After 4 hours, cells were centrifuged, viability dye was added (1:1000 dilution) and surface antibody staining was carried out (Table S1). After surface antibody staining and washing, cell pellets were resuspended in 150 μ l Cyto-Fast fixation/permeabilization working solution, prepared by adding fix/perm concentrate to fix/perm diluent according to manufacturer's instructions (Biolegend). Tubes were incubated for 30 minutes at 4°C in the dark. Without washing, 2ml of Cyto-fast Perm Wash solution (diluted to 1x with ddH₂O from 10x) was added to the tubes and cells were centrifuged at 400g for 5 minutes. Following washing, the cells were resuspended in 100 μ l intracellular staining mastermix with antibodies or isotype controls (Table S2) and incubated for 30 minutes at 4°C in the dark. The stained samples were then washed, fixed, and acquired. We calculated the alloreactive precursor frequency by adjusting the proliferation model in FlowJO. We calculated the precursor frequency by dividing the total number of divided alloreactive precursor T cells by the total number of all T cells (including alloreactive precursor T cells and non-dividing population, as previously described (Noorchashm et al. 1999).

Activation induced marker assay:

Cryopreserved patient PBMCs obtained before were resuspended in culture media (X-VIVO plus 10% AB Serum) at $2 \times 10^6 - 0.5 \times 10^6$ cells/ml and were co-cultured with a similar number of CFSE stained donor or third party splenocytes in the presence of 2 μ g/ml anti-CD40 blocking antibody (BioXcell). As donor stimulation, donor PBMCs isolated from deceased donor spleens were used and as third-party stimulation, PBMCs isolated from 3

different deceased 3rd Party donors were used. After 18 hours, cells were harvested, washed and cell surface staining was performed using fluorescent monoclonal antibodies directed against CD3, CD4, CD8, CD26, CD127, CD69, CD137, CD154, CD56, and GARP. Cells were then fixed and permeabilized using FOXP3/TF staining buffer kit (eBioscience) and stained using anti-FOXP3 and anti-Helios antibodies. Samples were immediately acquired on a BD LSRFortessa and data analyzed using FlowJo software. As a positive control for this assay, we stimulated recipient cells with 0.1 µg/mL Staphylococcus aureus enterotoxin B (SEB, Sigma-Aldrich). We excluded CFSE stained donor/third-party cells and alloantigen specific cells were defined as CD4⁺ T cells, CD8⁺ T cells and Helios⁺ Foxp3⁺ Tregs co-expressing CD154/CD69, CD137/CD69 and CD137/GARP, respectively. Experiments were conducted with one patient per run with all the time-points and with quality control ensured by including PBMCs from the same healthy volunteer in each run.

Table S1 – Flow cytometry panel for perfusate T cells subsets

Antibody	Fluorochrome	Detector	Clone	Cat. number	Volume (μ)
Viability	FITC	B530	Polyclonal	L23105	1/1000
CD4	BV605	V610	RM4-5	100547	2.5
CD45RA	BV650	V670	HI100	304135	2
CD62L	PE	YG586	DREG-56	304805	3
CD127	BV711	V710	HIL-7R-M2	563165	3
CD8	PE/Cy7	YG780	SK1	344711	1
CD3	APC-Cy7	R780	17A2	100221	1
CD25 (2A3)	APC	R670	M-A251	356109	5
CD25	APC	R670	BC96	302609	5

Table S2 – Flow cytometry panel for perfusate Monocytes, Dendritic cells, and B cell subsets

Antibody	Fluorochrome	Detector	Clone	Cat. Number	Volume(μl)
Viability	FITC	B530	Polyclonal	L23105	1/1000
CD123	BV421	V450	6H6	306017	1
CD38	V510	V525	HIT2	303539	1
CD24	V605	V610	M1/69	101827	1
CD27	BV711	V710	O323	302833	2
HLA-DR	BV785	V780	L243	307641	1
CD19	APC/CY7	R780	SJ25C1	363009	2
CD16	PerCP/Cy5.5	YG710	3G8	302027	1
CD3	PE	YG586	UCHT1	300407	1
CD11c	PE/Dazzle	YG610	N418	117347	1
CD14	PE/Cy7	YG780	M5E2	301813	2
IgD	APC	R670	IA6-2	348221	2

Table S3. Extracellular master mix for intracellular cytokine staining mix.

Antibody	Color	Volume (µl)
CD4 (Biolegend)	APC-CY7	3.5
CD8 (Biolegend)	PE-CY7	2
Responders (Added on D0) (e-Bioscience)	APC	1
Stimulants (Added on D0) (e-Bioscience)	V450	1
Viability (Invitrogen)	V450	1

Table S4. Intracellular antibodies and IGG controls for intracellular cytokine staining mix.

Isotype	Color	Volume (μl)
IgG1	PE	5
IgG	FITC	5
Antibody	Color	Volume (μl)
IFNγ	PE	5
Granzyme B	FITC	5

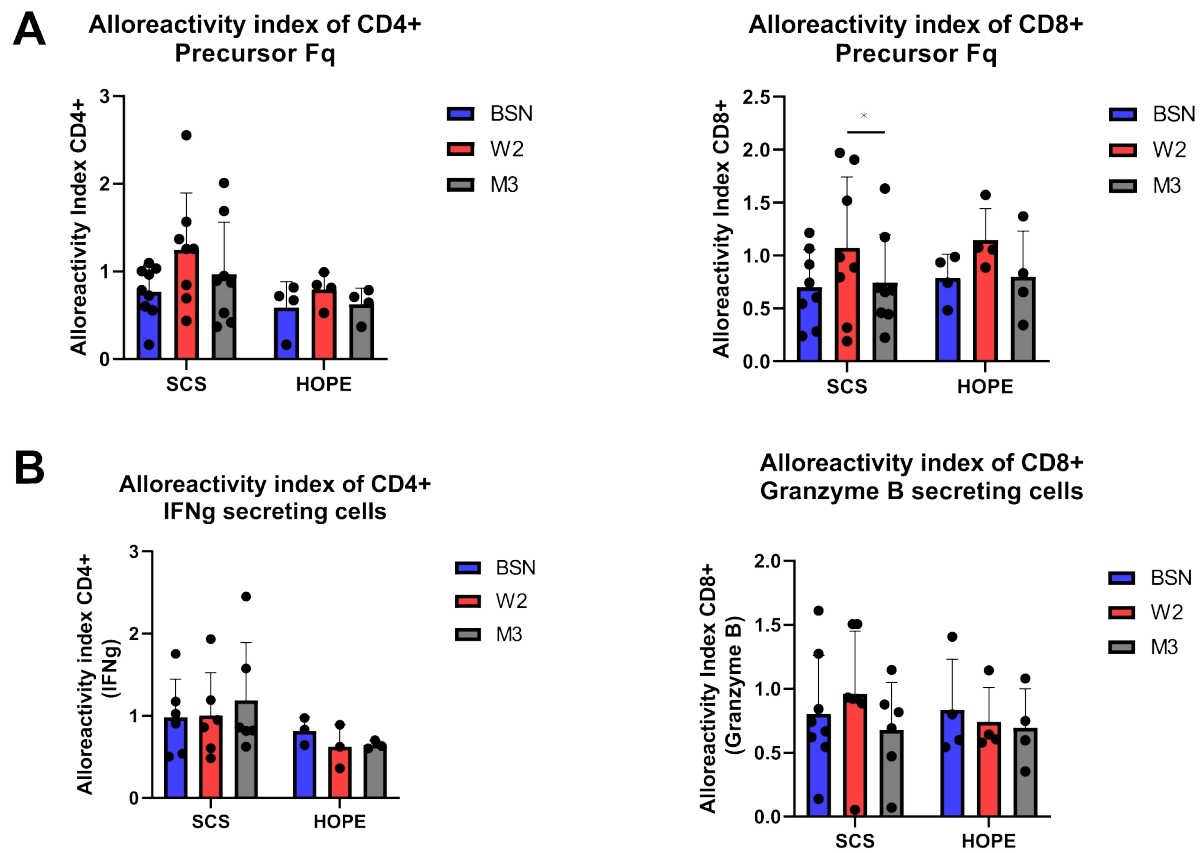


Figure S1. HOPE treated livers exhibit reduced alloreactive CD4+ and CD8+ cytokine secretion responses. (A) Quantification of alloreactive index for CD4+ Precursor frequency and CD8+ Precursor frequency in the two sample groups (SCS and HOPE) following a 5-day culture of recipient PBMCs with either Donor or Third-party allogenic cells. (B) Alloreactivity index calculated from intracellular cytokine staining of IFNg CD4+ secreting T cells and Granzyme B secreting CD8+ T cells following 5 days co-culture. Time-points include pre-transplantation baseline (BSN), week 2 following transplantation (W2) and month 3 following transplantation (M3). Each data point on the dot plot corresponds to an individual patient and represents an average of 3 replicates per experiment. Alloreactivity index was calculated as a ratio of donor-specific T cell response to third party T cell response. (A-B) Significance was determined by Friedman's test and p values represent pairwise comparisons of adjusted p values using Wilcoxon test. The values are shown as the mean $\pm$ SD. * $p < 0.05$

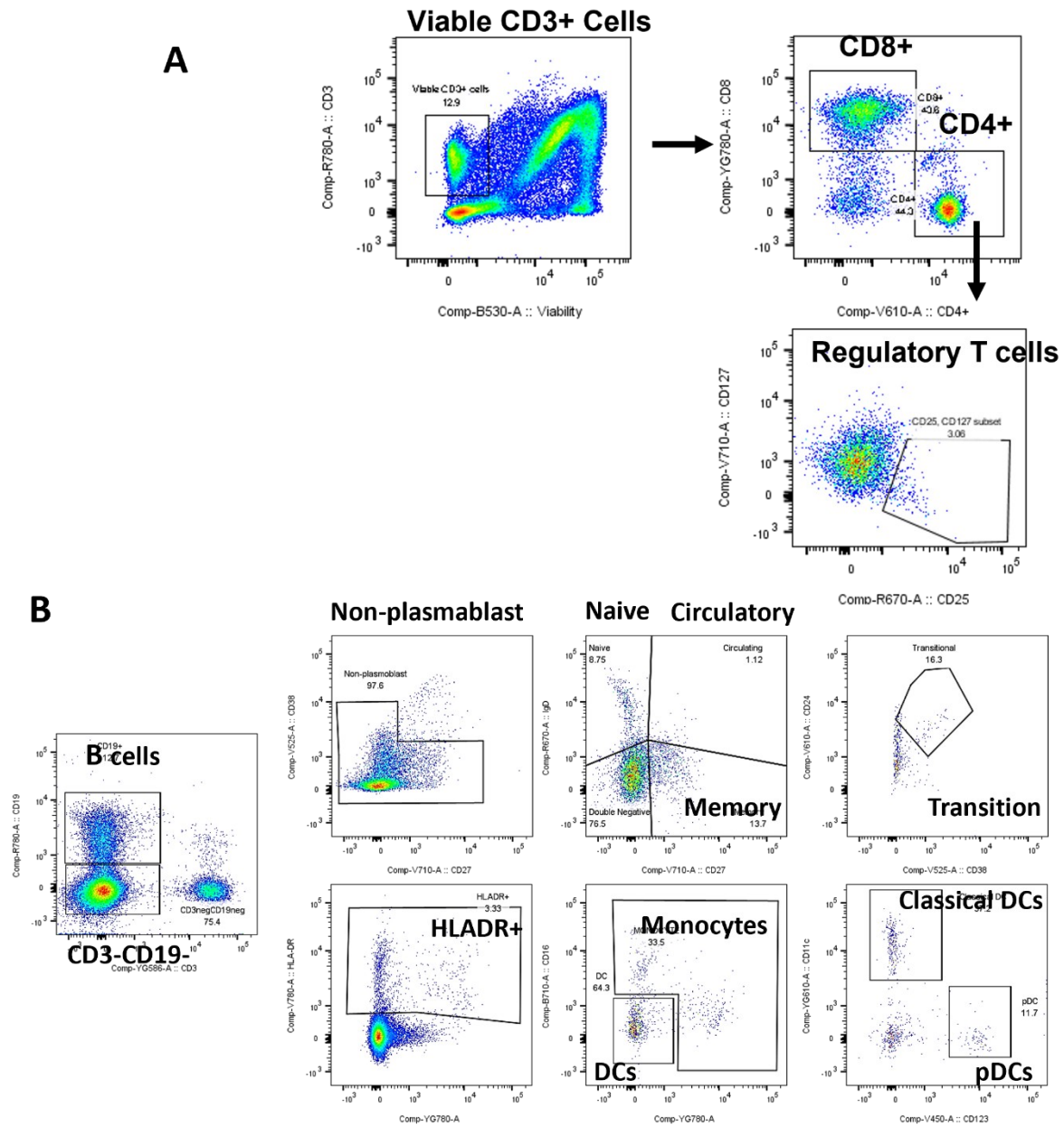


Figure S2. (A) gating strategy used to characterize CD8 and CD4 T cells and regulatory T cells based on the expression of CD3, CD8, CD4, CD25 and CD127. **(B)** gating strategy employed to identify monocytes and the different compartments of DCs and B cells.